👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.4

Describe Data Using the Mean

Lesson 2-8 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can determine the mean of a data set, use a target mean to find a missing value, and explain the mean as the fair share and the balance point of the data.

Language Objective: I can explain my reasoning using the words mean, average, measure of center, balance point, and data set.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean | Average | Measure of center | Balance point | Data set

☝️ *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The value found by adding all the data and dividing by how many values there are is the _____.

2. A common everyday word that means the same thing as the mean is the _____.

3. A single number that describes the typical value of a whole data set is a _____.

4. The value where the total amount above equals the total amount below is the _____.

5. The whole collection of collected values is the _____.

Write About the Math

How many points does Sofia need to score in the last game to hit her goal, and how does the balance-point view give the same answer as adding and dividing?

¿Cuántos puntos necesita anotar Sofía en el último partido para llegar a su meta, y cómo da la vista del punto de equilibrio la misma respuesta que sumar y dividir?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Mean** — *Media*
- ☐ **Average** — *Promedio*
- ☐ **Measure of center** — *Medida de centro*
- ☐ **Balance point** — *Punto de equilibrio*
- ☐ **Data set** — *Conjunto de datos*

Model: *A typical day is the amount Miguel would earn if he earned the SAME amount every day.* — Students say a typical day is the equal-share amount, and that sharing $60 across 4 days means dividing.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **After 9 games Sofia has scored ____ points. To average 15 across 10 games she needs ____ points in all, so game 10 must be ____ points. Using the balance point instead, her games total ____ points below the target mean and ____ points above it, so game 10 has to land ____ the target mean.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Average
3. **Notes 3:** Measure of center
4. **Notes 4:** Balance point
5. **Notes 5:** Data set
6. **Watch for this mistake:** *The most common mistake with the mean is adding correctly and then dividing by the wrong number — or not dividing at all. Students report the total (for 8, 10, 6, 12 they say the mean is 36) or they divide by a count that does not match the data, such as dividing a 4-value total by 5, or skipping a zero when counting the values. Before you finish, do two things: count the values in the data set out loud, and check that your mean lands between the smallest and largest value. A mean of 36 for data that never goes above 12 is impossible.*
7. **Try It 1:** C. 17 points — *She needs 150 points in all and already has 133, so game 10 must supply 150 − 133 = 17 points.*
8. **Try It 2:** C. 24 — *If every value moves up by 4, the balance point moves up by 4 as well, so the new mean is 20 + 4 = 24. In totals: adding 4 to each of n values adds 4n to the total, and 4n ÷ n = 4.*